

T13

Евронем: $n = 139$ $\sigma_g = 5,722$ $\sigma_u = 4,612$

Евронка: $m = 1000$ $\sigma_g = 6,161$ $\sigma_u = 3,055$

$L = 0,05$

Метрика: $\eta \sim N(\theta_1; \theta_2)$ Думка: $\xi \sim N(\theta_1; \theta_2)$

$H_0: \theta_{1eb} = \theta_{1er}$

$H_0: \theta_{2eb} = \theta_{2er}$

$H_1: \theta_{1eb} \neq \theta_{1er}$

$H_1: \theta_{2eb} \neq \theta_{2er}$

$S^2 = \frac{n}{n-1} \sigma^2$

$S_{g.er}^2 = 32,9785$ $S_{g.eb}^2 = 37,9959$ $S_{u.er}^2 = 21,4246$ $S_{u.eb}^2 = 25,5786$

$\tilde{D}_g = \frac{S_{g.er}^2}{S_{g.eb}^2} \approx 0,86795$ $\tilde{D}_u = \frac{S_{u.er}^2}{S_{u.eb}^2} \approx 0,8376$

$\tilde{D} \sim F(k-1, m-1) = F(138, 999) \rightarrow G: \tilde{D} \leq F_{\frac{\alpha}{2}}(138, 999) \text{ или } \tilde{D} \geq F_{1-\frac{\alpha}{2}}(138, 999)$

$F_{\frac{\alpha}{2}}(138, 999) \approx 0,76738$ $F_{1-\frac{\alpha}{2}}(138, 999) \approx 1,27169 \rightarrow \tilde{D}_u \text{ и } \tilde{D}_g \in G \rightarrow$

$W = P(\Delta \in G | H_1) = P(\tilde{D} \leq F_{\frac{\alpha}{2}}(138, 999) | H_1) + P(\tilde{D} \geq F_{1-\frac{\alpha}{2}}(138, 999) | H_1) =$

$= P\left(\frac{S_{er}^2}{S_{eb}^2} \cdot \frac{\sigma_{eb}^2}{\sigma_{er}^2} \leq F_{\frac{\alpha}{2}}(138, 999)\right) + P\left(\frac{S_{er}^2}{S_{eb}^2} \cdot \frac{\sigma_{eb}^2}{\sigma_{er}^2} \geq F_{1-\frac{\alpha}{2}}(138, 999)\right) \rightarrow W = \int_0^{F_{\frac{\alpha}{2}}(138, 999)} g(t) dt +$

$+ \int_{F_{1-\frac{\alpha}{2}}(138, 999)}^{\infty} g(t) dt$

0